# When Software Fabricates Assurance: Seven Design Principles for Trustworthy Governance, Risk and Compliance Systems

**Name of the Track:** Track 1 – Information Systems: Theory and Practice (Digital Ethics and Policy: Regulation and Compliance)

**Type of Submission:** Ongoing research (design science research; artifact and technical evaluation completed, naturalistic evaluation pending)


## Abstract

### Purpose

Boards, auditors and regulators rely on governance, risk and compliance (GRC) information systems for evidence that risks are identified, rated, approved and controlled. Research on these systems has examined their adoption and value. We found no study asking whether the evidence they present can itself be trusted. A code and runtime audit of an existing risk management application found that the software invented risk ratings, discarded approvals, accepted unauthenticated authority and reported failed database writes as successes. This study asks which design failures compromise the integrity of assurance evidence. It also asks which design principles prevent them, and how well an instantiation of those principles performs against the original system.

### Design/Methodology/Approach

The study follows design science research. The audit produced a taxonomy of seven failure modes. Each was matched to a design principle grounded in established theory: integrity models, fail-safe defaults, complete mediation, separation of duties, tamper-evident logging and the critique of ritualised assurance. The principles were instantiated in a new system, ERMA-GRC. Both systems were then run against identical stored data in 23 scenarios, whose protocol was time-stamped in the project repository before execution. Eighteen scenarios derive from the audited failures, four test the principles beyond them, and one was expected to fail. Every deviation from the protocol is reported.

### Findings

The original system prevented the failure in 4 of the 18 audit-derived scenarios; ERMA-GRC prevented it in all 18. ERMA-GRC accepted no unauthenticated writes, rejected self-approval, recorded audit entries only under the signed-in identity, detected an edited audit entry and returned errors instead of false successes. It also prevented 3 of the 4 failures in the scenarios beyond the audit, including a race between two simultaneous approval decisions, and every stored-state outcome was read from the database stand-in, not from what either system reported. Two failures of the new design are reported with equal weight. First, a key risk indicator's status was computed correctly in the interface but could be declared by a client calling the server directly. Second, an insider holding the audit key could rewrite the evidence trail undetected after a restart. The principles also depend on one another: separation of duties relies on the integrity of the audit trail, and a rule enforced only in the client offers no assurance.

**Originality/Value**

The study treats assurance integrity as a design property of GRC information systems rather than an organisational outcome. It contributes three things: a failure taxonomy derived from a real system, seven design principles with two testable propositions toward a nascent design theory, and a reusable scenario protocol. Auditors and buyers can use the principles as a review checklist. The documented boundaries point to future work: externally anchored audit trails, server-side computation of every assurance metric, and evaluation with practitioners.

**Keywords:** assurance integrity; governance, risk and compliance; design science research; enterprise risk management; audit trail; separation of duties

## 1. Introduction

Organisations increasingly record their risk management in governance, risk and compliance (GRC) information systems. A risk register, a heatmap, an approved assessment and an audit trail are the evidence that boards, auditors and regulators rely on to judge whether risks are controlled (Power, 1997). Such a system is valuable only if what it shows is true.

This study began with a code and runtime audit of an existing GRC application documented as production-ready. The software produced assurance evidence with no underlying record. It invented ratings for unrated risks from record identifiers or default loss values, reset every approval to draft on the next save, kept an "immutable" audit trail in the browser where anyone could rewrite it, and reported failed database writes as successes. These were ordinary design decisions, not exotic security flaws, and each turned the system into a source of risk rather than a control over it.

Within the literature reached by our search, this problem is unaddressed. GRC information systems research has studied adoption and value (Papazafeiropoulou & Spanaki, 2016; Racz et al., 2010), enterprise risk management research warns that assurance can decay into ritual (Power, 2009), and the relevant mechanisms are understood in generic form (Clark & Wilson, 1987; Saltzer & Schroeder, 1975). We found no study connecting application design failures to corrupted assurance evidence or testing design principles against a real system.

We therefore ask which design failures in a GRC information system compromise the integrity of its assurance evidence (RQ1); which theory-grounded design principles address them (RQ2); and to what extent an instantiation of these principles prevents the failures compared with the original system (RQ3). Using design science research, we contribute a failure taxonomy, seven design principles and a time-stamped scenario evaluation of the artifact, ERMA-GRC. Sections 2 to 6 present the literature, method, results, implications and limitations.

## 2. Literature Review

### 2.1 Conceptual Background

*Enterprise risk management* (ERM) is the organisation-wide identification, assessment, treatment and monitoring of risk; *governance, risk and compliance (GRC) information systems* integrate these activities and keep their records (Racz et al., 2010). Those records give boards, auditors and regulators *assurance* that risks are known and controlled. We define *assurance integrity* as the property that every assurance item a GRC system presents (a rating, an approval, an exposure figure or an audit entry) is either a recorded, attributable action or an explicit statement that no such record exists. It is a property of the system's design, not of the risks it manages.

### 2.2 Review of Previous Studies

Organisational research shows that ERM is enacted rather than adopted. Arena et al. (2010) trace how risk rationalities, experts and technologies combine differently across firms; Mikes (2009) identifies distinct "calculative cultures"; Mikes and Kaplan (2015) argue that a single ERM template is itself a risk; and Meidell and Kaarbøe (2017) show that a risk function's tools shape its influence. Power (1997) warns that verification rituals can yield auditability rather than assurance, and Power (2009) calls ERM at its weakest a "risk management of nothing". Technology therefore helps produce risk, so its design merits study.

GRC information systems research has defined the field (Racz et al., 2010), identified implementation success factors (Papazafeiropoulou & Spanaki, 2016), and described an adoption lifecycle and value drivers (Spanaki & Papazafeiropoulou, 2016). It asks how these systems are adopted and what value they deliver, not whether their evidence can be trusted.

The artifact's mechanisms exist in generic form: well-formed transactions, integrity verification and separation of duty (Clark & Wilson, 1987); fail-safe defaults and complete mediation (Saltzer & Schroeder, 1975); separation of duties in workflows (Botha & Eloff, 2001); and tamper-evident logs, using hash chains with evolving keys (Schneier & Kelsey, 1999) or history trees kept honest by external auditors (Crosby & Wallach, 2009). Continuous assurance monitors business-process controls through analytics (Alles et al., 2006; Vasarhelyi et al., 2004), focusing on monitored processes rather than faults in the monitoring software. Table 1 summarises the seven closest studies; five were assessed from abstracts only.

**Table 1. Summary of Selected Literature**

| Author(s) | Context | Methodology | Key Findings | Gap Identified |
|---|---|---|---|---|
| Racz et al. (2010) | Integrated GRC | Literature analysis, definition | A frame of reference for GRC research | Not its focus: artifact design and evaluation |
| Papazafeiropoulou & Spanaki (2016) | GRC IS implementation | Literature review and expert interviews | Implementation characteristics and success factors | Integrity of the evidence the systems produce |
| Spanaki & Papazafeiropoulou (2016) | GRC IS adoption | Qualitative study | Adoption lifecycle; integration, optimisation and information as value drivers | Not its focus: trustworthiness of that information |
| Arena et al. (2010) | ERM in organisations | Comparative case study | ERM varies with its rationalities, experts and technologies | Not its focus: how a technology's design distorts risk information |
| Alles et al. (2006) | Continuous control monitoring | Pilot implementation (Siemens) | Business-process controls can be monitored continuously | Not its focus: design faults in the monitoring software |

| Author(s) | Context | Methodology | Key Findings | Gap Identified |
|---|---|---|---|---|
| Botha & Eloff (2001) | Workflow access control | Conceptual model | Static and dynamic separation-of-duties enforcement | Not its focus: approval semantics of risk assessments |
| Crosby & Wallach (2009) | Secure logging | Data structure and analysis | Efficient tamper-evident logs, provided auditors hold commitments | Business workflows and other failure types |

### 2.3 Research Gap

Within the web-indexed literature our search reached, with every DOI verified, no study examines how a GRC application's design can itself corrupt the assurance evidence it presents: by fabricating values, overwriting decisions, accepting unauthenticated authority or reporting failures as success. GRC research stops at adoption and value, while mechanism research does not address the evidence that risk and control self-assessment produces: ratings, approvals, appetite exposure and indicator status. The gap is therefore not a missing mechanism but missing design knowledge that maps application failures to corrupted evidence and tests principles against a real system.

### 2.4 Research Framework

The study links observed failure modes (RQ1), theory-grounded principles (RQ2), their mechanisms, and scenario outcomes testing whether the failures are prevented (RQ3). These contribute components of a design theory (Gregor & Jones, 2007): purpose and scope, constructs, principles of form and function, and testable propositions (Section 5). Each principle pairs an aim with a mechanism and a kernel-theory rationale (Gregor et al., 2020), listed in Table 2.

## 3. Methodology

### 3.1 Research Design

We followed design science research (DSR; Hevner et al., 2004), entering the design science research methodology (DSRM; Peffers et al., 2007) at problem identification. Its six activities were (1) problem identification, the audit (RQ1); (2) objectives; (3) design of seven principles, instantiated in ERMA-GRC (RQ2); (4) demonstration on Oracle Autonomous Database; (5) evaluation (RQ3); and (6) communication. The evaluation is one artificial, summative episode of the Technical Risk and Efficacy strategy of the Framework for Evaluation in Design Science (FEDS; Venable et al., 2016).

### 3.2 Data Collection

Evidence came from the audit and a scenario experiment, its protocol time-stamped in the project repository (not a public registry) before execution. The original system (baseline) and ERMA-GRC ran against one database stand-in with identical rows (Figure 1), one recorded run per scenario, keeping full evidence.

### 3.3 Measures and Instruments

Eighteen of the 23 scenarios operationalise the failure modes, four test the principles beyond them, and one was expected to fail. Runs were coded PREVENTED, OCCURRED or not applicable against criteria registered in advance, reading stored state from the stand-in and display outcomes from screen text.

### 3.4 Data Analysis

We count PREVENTED scenarios per failure mode; deterministic outcomes need no inferential statistics. All deviations are reported in a replication package available on request: the test scripts were piloted after registration; open points (e.g., an approver as editor in S2.2) were fixed before each run; the rule coding G4 by its stricter path was not registered; and S5.1 was re-run after a script fault.

**Use of generative AI.** Claude (Claude Opus 5.5, Anthropic) was used under the authors' direction to assist the audit, implement much of ERMA-GRC, draft the protocol, write and run the test scripts, propose outcome codings, produce the figures, support the literature search, and draft text. The authors set the questions, approved each stage including the protocol before execution, verified sources and results, revised the text and take full responsibility. No AI tool is an author.

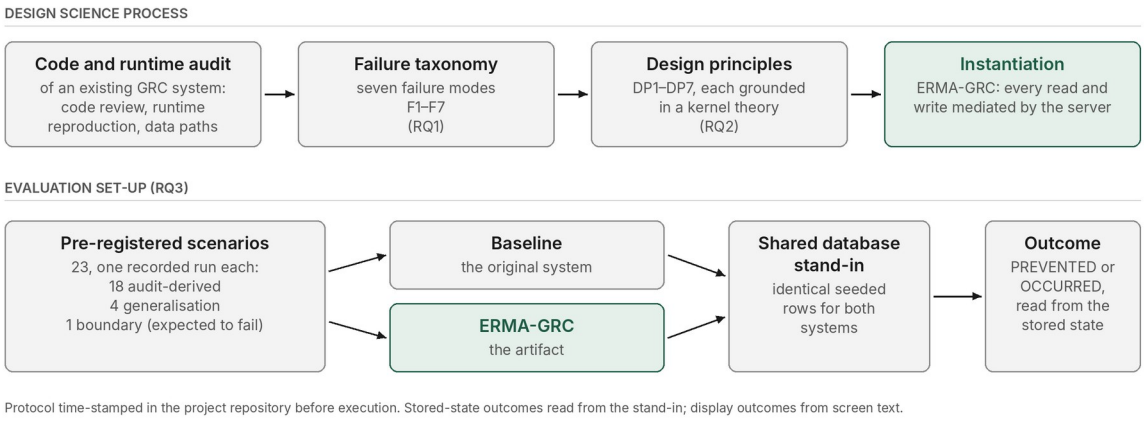



**Figure 1.** DSR process and evaluation set-up: the audit yields F1–F7, which inform DP1–DP7, instantiated in ERMA-GRC; both systems run the 23 scenarios against one database stand-in.

## 4. Results and Discussion

### 4.1 Results

**RQ1 and RQ2.** Table 2 maps each failure mode to its principle, kernel theory and mechanism in ERMA-GRC.

**Table 2. Failure Modes, Design Principles, Kernel Theories and Mechanisms**

| Failure mode (RQ1) | Design principle (RQ2) | Kernel theory | Mechanism in ERMA-GRC |
|---|---|---|---|
| F1 Fabricated assurance values | DP1 Evidence provenance | Clark & Wilson (1987); Power (1997) | No derived ratings; "Not rated" shown |

| Failure mode (RQ1) | Design principle (RQ2) | Kernel theory | Mechanism in ERMA-GRC |
|---|---|---|---|
| F2 Decisions overwritten by defaults | DP2 Decision preservation | Clark & Wilson (1987); Saltzer & Schroeder (1975) | Defaults only at creation; approvals only via authorised transitions |
| F3 Client-side or absent enforcement | DP3 Server-enforced, identity-bound separation of duties | Saltzer & Schroeder (1975); Botha & Eloff (2001) | Server session; deny by default; maker ≠ checker; conflicts refused; revocation |
| F4 Mutable, forgeable evidence trail | DP4 Tamper-evident, actor-bound trail | Schneier & Kelsey (1999); Crosby & Wallach (2009) | Keyed hash chain (static key); session-bound actor; visible verification |
| F5 Silent failure reported as success | DP5 Fail-visible operation | Saltzer & Schroeder (1975) | Errors returned; no fallback store |
| F6 Fragmented system of record | DP6 Single system of record | Clark & Wilson (1987) | Records in database; accounts and audit chain on server |
| F7 Destructive actions on fabricated records | DP1 Evidence provenance | as for F1 | Empty states, no placeholder rows |
| F1 (metrics) | DP7 Computed, not declared, metrics | Vasarhelyi et al. (2004); Cox (2008) | Status computed from thresholds, in the client |

**RQ3.** ERMA-GRC prevented all 18 audit-derived failures; the baseline prevented 4 (Table 3). The baseline stored DRAFT after an approver set APPROVED, accepted writes without a session or with a self-declared administrator header, recorded an audit entry under a claimed "CEO" identity, reported success while the database was unreachable, and invented two scored risks, each with a delete button, for a cycle that had none. ERMA-GRC prevented three of the four generalisation failures: it kept one of two simultaneous approval decisions, rejected a write from a deactivated user's still-valid session, and rejected a path-traversal identifier before any database request. The baseline's 0/4 there largely repeats its F2 and F3 defects and is not independent evidence. For balance, when an editor without approval rights changed an approved risk (S2.2 variant), ERMA-GRC returned it to review by design, which the literal criterion codes as OCCURRED; the simultaneous-decision race on a record's status (G1 variant) was prevented.

**Table 3. Scenario Outcomes (PREVENTED / Runnable)**

| Scenarios | Baseline | ERMA-GRC |
|---|---|---|
| F1 Fabricated values (S1.1–S1.4) | 3 / 4 | 4 / 4 |
| F2 Decisions overwritten (S2.1–S2.2) | 0 / 2 | 2 / 2 |
| F3 Enforcement (S3.1–S3.4) | 0 / 4 | 4 / 4 |
| F4 Evidence trail (S4.1–S4.3) | 1 / 3 | 3 / 3 |
| F5 Silent failure (S5.1–S5.3) | 0 / 3 | 3 / 3 |

| Scenarios | Baseline | ERMA-GRC |
|---|---|---|
| F6 Fragmented record (S6.1) | 0 / 1 | 1 / 1 |
| F7 Fabricated-row deletion (S7.1) | 0 / 1 | 1 / 1 |
| Generalisation (G1–G4) | 0 / 4 | 3 / 4 |
| Boundary (L1, expected to fail) | 0 / 1 | 0 / 1 |
| **Total** | **4 / 23** | **21 / 23** |

**Where the baseline passed.** The baseline displayed a stored statement and single loss expectancy (SLE) faithfully (S1.2), showed a missing approval status as "Draft", as ERMA-GRC does, rather than Approved (S1.3), held no invented entries in a fresh audit trail (S1.4), and attributed an audit entry to the self-asserted acting persona (S4.3). S1.3 and S1.4 contradict the audit report; for S1.4, the seeding code exists but never runs on a fresh installation.

**Where ERMA-GRC failed.** In G4, a key risk indicator (KRI) reading received its status by rule in the interface, but the API stored a status declared by the client, as the baseline's does. In L1, an insider holding the audit key re-chained an edited log offline, and verification passed after a restart. Figure 2 contrasts two baseline failures with ERMA-GRC on the same stored data.

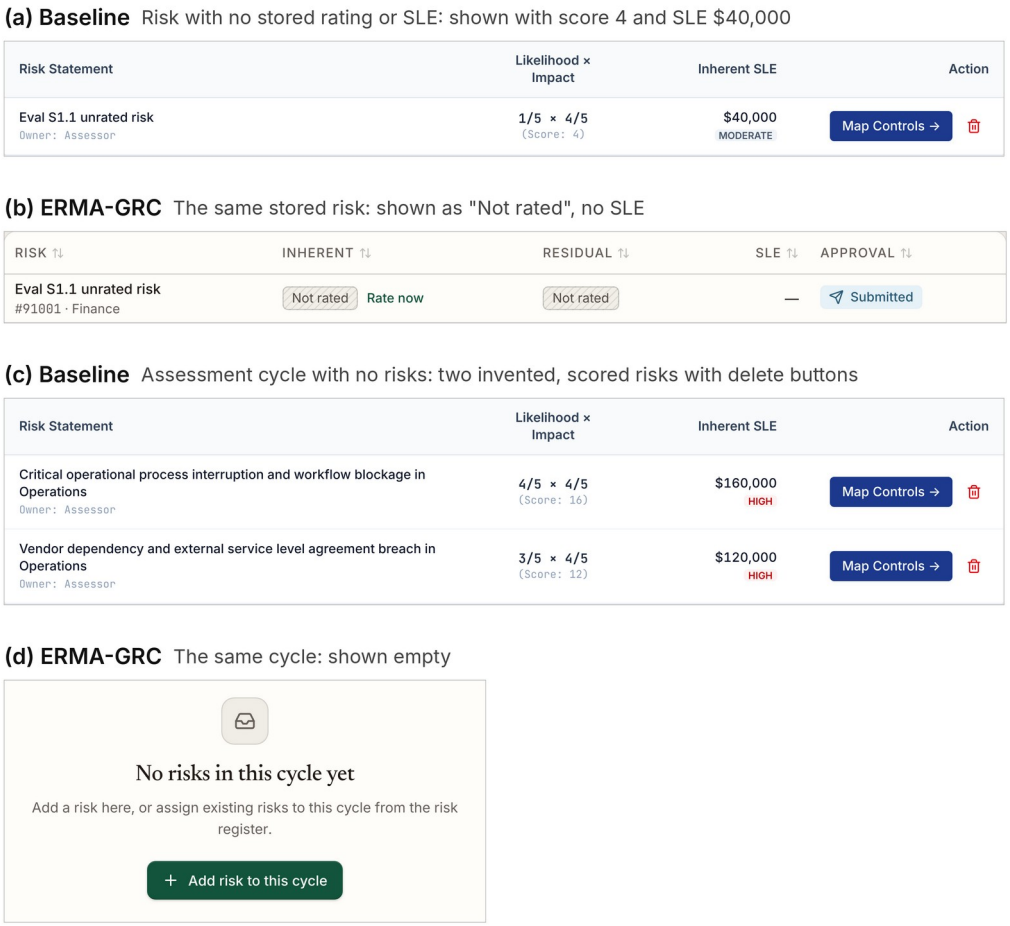



**Figure 2.** The same stored data in both systems: (a) the baseline shows an unrated risk with score 4 and SLE $40,000 (S1.1); (b) ERMA-GRC shows it as "Not rated"; (c) the baseline

invents two scored, deletable risks for an empty cycle (S7.1); (d) ERMA-GRC shows the cycle as empty. Illustrative post-evaluation re-captures; unrelated columns hidden.

### 4.2 Discussion

**The failures are assurance failures.** The baseline produced scores, approvals and audit entries disconnected from its stored state: a software-made "risk management of nothing" (Power, 2009). Established mechanisms counter this: fail-safe defaults and complete mediation (Saltzer & Schroeder, 1975), separation of duties (Botha & Eloff, 2001), and well-formed transactions with integrity verification (Clark & Wilson, 1987). DP1, DP2 and DP7 specialise these integrity ideas to assurance evidence: they ask whether a displayed rating, decision or metric is evidence at all.

**The principles depend on each other.** ERMA-GRC reads a record's creator, and hence who may not approve it, from its audit trail; in L1, rewriting one creation entry changed who was barred from approving that record. DP3 therefore rests on DP4, and DP4 on keys held outside the application. Unlike Schneier and Kelsey's (1999) evolving keys, ERMA-GRC's key is static, and without external commitments its holder can rewrite history (Crosby & Wallach, 2009).

**Enforcement must sit at the server.** G4 shows that a rule checked only in the client is a convention, not a control. Integrity is necessary but not sufficient: faithful records of a poorly calibrated risk matrix are still poor risk information (Cox, 2008).

## 5. Implications

**Theoretical implications.** The study recasts assurance integrity as a design property of GRC systems and contributes components of a nascent design theory (Gregor & Jones, 2007): purpose and scope, constructs, seven principles and two testable propositions. P1: separation of duties is only as strong as the integrity of the evidence trail it reads. P2: an assurance rule enforced only in the client can be defeated by any authenticated client.

**Practical implications.** Before purchase, buyers and auditors can test whether an unrated risk is shown as unrated, whether an ordinary save can overwrite an approval, whether authority comes from a verified session, whether the audit trail is verifiable and anchored outside the vendor's control, and whether a failed write can ever appear as a success.

**Societal implications.** Regulators and the public increasingly rely on compliance claims produced by software. Systems that cannot fabricate evidence make those claims more trustworthy and accountability harder to evade.

## 6. Conclusion, Limitations and Future Research

This study asked how a GRC system's design can corrupt its assurance evidence, and how to prevent it. An audit yielded seven failure modes, addressed by seven theory-grounded principles. ERMA-GRC, which instantiates them, prevented every audit-derived failure; the original system prevented four of eighteen. Its two failures, a metric enforced only in the client and an audit trail its key holder can rewrite, are part of the contribution.

**Limitations.** The evidence comes from one case. The authors built the artifact and wrote the scenarios with an AI assistant involved in every stage, from audit to outcome coding; time-stamping the protocol and reporting every outcome reduce but do not remove this bias. The evaluation was artificial: both systems ran against a stand-in derived from ERMA-GRC's own test double, so concurrency and outage results depend on its fidelity, and the display

scenarios lacked a rated risk as a positive control. The literature search lacked Scopus and Web of Science, and no commercial product was compared.

**Future research.** Next steps are naturalistic evaluation with practitioners, multiple-case studies across GRC products, externally anchored audit commitments, server-side computation of every assurance metric, and longitudinal study of how integrity-preserving systems change the use of risk information.

## References


Alles, M., Brennan, G., Kogan, A., & Vasarhelyi, M. A. (2006). Continuous monitoring of business process controls: A pilot implementation of a continuous auditing system at Siemens. *International Journal of Accounting Information Systems, 7*(2), 137–161. https://doi.org/10.1016/j.accinf.2005.10.004

Arena, M., Arnaboldi, M., & Azzone, G. (2010). The organizational dynamics of enterprise risk management. *Accounting, Organizations and Society, 35*(7), 659–675. https://doi.org/10.1016/j.aos.2010.07.003

Botha, R. A., & Eloff, J. H. P. (2001). Separation of duties for access control enforcement in workflow environments. *IBM Systems Journal, 40*(3), 666–682. https://doi.org/10.1147/sj.403.0666

Clark, D. D., & Wilson, D. R. (1987). A comparison of commercial and military computer security policies. In *Proceedings of the 1987 IEEE Symposium on Security and Privacy* (pp. 184–194). IEEE. https://doi.org/10.1109/SP.1987.10001

Cox, L. A., Jr. (2008). What's wrong with risk matrices? *Risk Analysis, 28*(2), 497–512. https://doi.org/10.1111/j.1539-6924.2008.01030.x

Crosby, S. A., & Wallach, D. S. (2009). Efficient data structures for tamper-evident logging. In *Proceedings of the 18th USENIX Security Symposium* (pp. 317–334). USENIX Association. https://www.usenix.org/legacy/event/sec09/tech/full_papers/crosby.pdf

Gregor, S., Chandra Kruse, L., & Seidel, S. (2020). Research perspectives: The anatomy of a design principle. *Journal of the Association for Information Systems, 21*(6), 1622–1652. https://doi.org/10.17705/1jais.00649

Gregor, S., & Jones, D. (2007). The anatomy of a design theory. *Journal of the Association for Information Systems, 8*(5), 312–335. https://doi.org/10.17705/1jais.00129

Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. *MIS Quarterly, 28*(1), 75–106. https://doi.org/10.2307/25148625

Meidell, A., & Kaarbøe, K. (2017). How the enterprise risk management function influences decision-making in the organization – A field study of a large, global oil and gas company. *The British Accounting Review, 49*(1), 39–55. https://doi.org/10.1016/j.bar.2016.10.005

Mikes, A. (2009). Risk management and calculative cultures. *Management Accounting Research, 20*(1), 18–40. https://doi.org/10.1016/j.mar.2008.10.005

Mikes, A., & Kaplan, R. S. (2015). When one size doesn't fit all: Evolving directions in the research and practice of enterprise risk management. *Journal of Applied Corporate Finance, 27*(1), 37–40. https://doi.org/10.1111/jacf.12102

Papazafeiropoulou, A., & Spanaki, K. (2016). Understanding governance, risk and compliance information systems (GRC IS): The experts view. *Information Systems Frontiers, 18*(6), 1251–1263. https://doi.org/10.1007/s10796-015-9572-3

Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of Management Information Systems, 24*(3), 45–77. https://doi.org/10.2753/MIS0742-1222240302

Power, M. (1997). *The audit society: Rituals of verification*. Oxford University Press.

Power, M. (2009). The risk management of nothing. *Accounting, Organizations and Society, 34*(6–7), 849–855. https://doi.org/10.1016/j.aos.2009.06.001

Racz, N., Weippl, E., & Seufert, A. (2010). A frame of reference for research of integrated governance, risk and compliance (GRC). In B. De Decker & I. Schaumüller-Bichl (Eds.), *Communications and multimedia security* (Lecture Notes in Computer Science, Vol. 6109, pp. 106–117). Springer. https://doi.org/10.1007/978-3-642-13241-4_11

Saltzer, J. H., & Schroeder, M. D. (1975). The protection of information in computer systems. *Proceedings of the IEEE, 63*(9), 1278–1308. https://doi.org/10.1109/PROC.1975.9939

Schneier, B., & Kelsey, J. (1999). Secure audit logs to support computer forensics. *ACM Transactions on Information and System Security, 2*(2), 159–176. https://doi.org/10.1145/317087.317089

Spanaki, K., & Papazafeiropoulou, A. (2016). The implementation of governance, risk, and compliance IS: Adoption lifecycle and enterprise value. *Information Systems Management, 33*(4), 302–315. https://doi.org/10.1080/10580530.2016.1220214

Vasarhelyi, M. A., Alles, M. G., & Kogan, A. (2004). Principles of analytic monitoring for continuous assurance. *Journal of Emerging Technologies in Accounting, 1*(1), 1–21. https://doi.org/10.2308/jeta.2004.1.1.1

Venable, J., Pries-Heje, J., & Baskerville, R. (2016). FEDS: A framework for evaluation in design science research. *European Journal of Information Systems, 25*(1), 77–89. https://doi.org/10.1057/ejis.2014.36